EWB AWS INTERNSHIP PROJECT REPORT

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Batch: 1

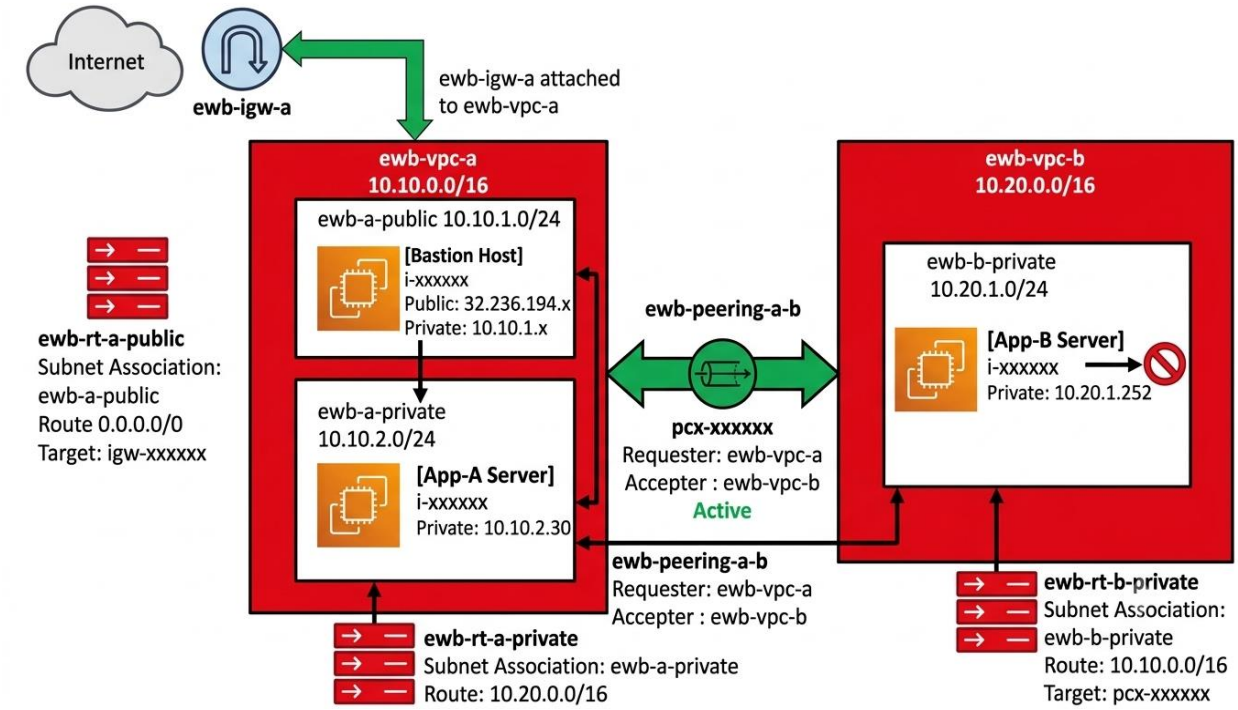
Project Title: AWS VPC Peering

Region: ap-southeast-2 - Asia Pacific / Sydney

Completion Date: October 20, 2026

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1. Architecture Diagram



2. Resource Inventory & Network Parameters

VPC CIDR Blocks: VPC-A (ewb-vpc-a): 10.10.0.0/16 | VPC-B (ewb-vpc-b): 10.20.0.0/16

Subnet CIDR Ranges: ewb-a-public (10.10.1.0/24), ewb-a-private (10.10.2.0/24), ewb-b-private (10.20.1.0/24)

Bastion Host (ewb-ec2-bastion): Public IP: 32.236.194.219 | Private IP: 10.10.1.x | Subnet: ewb-a-public

App-A Server (ewb-ec2-app-a): Private IP: 10.10.2.30 | Subnet: ewb-a-private | Security Group: ewb-sg-app-a

App-B Server (ewb-ec2-app-b): Private IP: 10.20.1.252 | Subnet: ewb-b-private | Security Group: ewb-sg-app-b

VPC Peering Connection: pcx-xxxxxx (Requester: ewb-vpc-a, Accepter: ewb-vpc-b, Status: Active)

Internet Gateway: ewb-igw-a (Attached to ewb-vpc-a)

3. Route Tables & Subnet Associations

Route Table 1: ewb-rt-a-public (Public Route Table)

Associated Subnet: ewb-a-public (10.10.1.0/24)

Routes: 10.10.0.0/16 -> Local | 0.0.0.0/0 -> ewb-igw-a (Internet Gateway)

Route Table 2: ewb-rt-a-private (App-A Private Route Table)

Associated Subnet: ewb-a-private (10.10.2.0/24)

Routes: 10.10.0.0/16 -> Local | 10.20.0.0/16 -> pcx-xxxxxx (VPC Peering Connection)

Route Table 3: ewb-rt-b-private (App-B Private Route Table)

Associated Subnet: ewb-b-private (10.20.1.0/24)

Routes: 10.20.0.0/16 -> Local | 10.10.0.0/16 -> pcx-xxxxxx (VPC Peering Connection)

4. Security Groups & Inbound Rules

Bastion Security Group (ewb-sg-bastion)

Rules: Inbound: SSH (Port 22) from Source 0.0.0.0/0 (My IP) for admin terminal access.

App-A Security Group (ewb-sg-app-a)

Rules: Inbound Rule 1: SSH (Port 22) from Source 10.10.1.0/24 (Bastion Subnet).
Inbound Rule 2: All ICMP - IPv4 from Source 10.20.0.0/16 (VPC-B Range).

App-B Security Group (ewb-sg-app-b)

Rules: Inbound Rule 1: SSH (Port 22) from Source 10.10.0.0/16 (VPC-A Range).
Inbound Rule 2: All ICMP - IPv4 from Source 10.10.0.0/16 (VPC-A Range).

5. Forward & Return Explanation

Forward Path Flow: App-A (10.10.2.30) sends an ICMP Echo Request to App-B (10.20.1.252). The route table (ewb-rt-a-private) routes target 10.20.0.0/16 to the peering connection (pcx-xxxxxx). Traffic crosses the AWS backbone into VPC-B, passes ewb-rt-b-private, and is allowed by App-B's security group (ewb-sg-app-b) ICMP inbound rule.

Return Path Flow: App-B responds with an ICMP Echo Reply to 10.10.2.30. VPC-B's route table (ewb-rt-b-private) directs target 10.10.0.0/16 back over pcx-xxxxxx to VPC-A. The reply evaluates against App-A's security group (ewb-sg-app-a) statefully and reaches App-A, achieving 0% packet loss.

6. Test Evidence & Validation Records

Test 1: Baseline Failure Proof (Initial ICMP Block)

Host: ec2-user@ip-10-10-2-30

Test Command: ping 10.20.1.252 --- Prior to adding ICMP inbound rule

Result: 100% packet loss.

Explanation: The baseline test confirmed that even though VPC Peering routes were active, App-B's firewall blocked the ICMP Echo Request, demonstrating that network routing alone is insufficient without proper security group authorization.

Test 2: Forward & Reverse Ping Success Proof

Host: ec2-user@ip-10-10-2-30

Test Command: ping 10.20.1.252 --- After adding All ICMP IPv4 rule

Result: 7 packets transmitted, 7 received, 0% packet loss.

Explanation: Adding the ICMP inbound rule allowed full round-trip communication across the peering tunnel, confirming 0% packet loss and verified forward and return path operation.

Test 3: App-B SSH Proof & Key Management

Host: ec2-user@ip-10-10-1-x (Bastion Host) -> ec2-user@ip-10-10-2-30 (App-A)

Test Command: chmod 400 bastion-key.pem && ssh -i bastion-key.pem ec2-user@10.10.2.30

Result: Successful SSH login

Explanation: Permissions were adjusted to 400 (read-only by owner) on bastion-key.pem to satisfy SSH client security requirements, enabling secure jump-box access to private instances.

Test 4: Controlled ICMP Failure & Recovery Validation

Summary: To validate firewall state enforcement, the ICMP inbound rule was temporarily removed from ewb-sg-app-b. Re-running ping 10.20.1.252 immediately resulted in 100% packet loss . Upon re-configure the ICMP inbound rule ping traffic resumed instantly with 0% packet loss

7. Cleanup Evidence

1. Terminated all EC2 instances.
2. Deleted VPC Peering Connection.
3. Detached from ewb-vpc-a. and Deleted Internet Gateway ewb-igw-a .
4. Deleted custom VPCs ewb-vpc-a and ewb-vpc-b along with associated subnets and route tables to prevent unauthorized billing.

Summary

Through this hands-on project, I gain knowledge inn multi-VPC architecture design, CIDR block allocation without overlapping ranges, bastion jump host configuration, SSH key permission security and bidirectional route table propagation over VPC Peering. Also gained practical experience troubleshooting network ,security group firewall rules.